Notes Title

Author Name

September 2, 2026

Abstract

These notes develop a small piece of theoretical machinery and work through its consequences in detail. The exposition is kept self-contained, with definitions, statements, short proofs, and worked examples interleaved with prose. The closing sections collect a few exercises for practice.

Contents

1 Setting

These notes develop a small piece of theoretical machinery to illustrate the structure of the template. The exposition follows the typical pattern of mathematical and physics notes: definitions, statements of results, short proofs, and worked examples interleaved with prose.

Citations are rendered in brackets, e.g. [1], and multiple sources combine and compress to ranges automatically [1, 2].

2 Definitions and Statements

Definition 2.1. Let $\mathcal{H}$ be a complex separable Hilbert space. A *state* is a unit vector $\psi \in \mathcal{H}$, considered up to a phase factor $e^{i\theta}$.

Definition 2.2. An *observable* is a self-adjoint linear operator $\hat{A}$ acting on $\mathcal{H}$.

The expectation value of an observable in a given state is defined by

$$\langle A \rangle_\psi = \langle \psi | \hat{A} | \psi \rangle. \tag{1}$$

Real expectation values

Theorem 2.3. *Expectation values of self-adjoint operators are real.*

Proof. By definition, $\overline{\langle A \rangle_\psi} = \overline{\langle \psi | \hat{A} | \psi \rangle} = \langle \psi | \hat{A}^\dagger | \psi \rangle = \langle \psi | \hat{A} | \psi \rangle = \langle A \rangle_\psi$, where we used $\hat{A}^\dagger = \hat{A}$. □

Remark 2.4. The reality of expectation values is what makes self-adjoint operators the natural mathematical model of observable quantities in quantum mechanics.

2.1 Time Evolution

Time evolution in quantum mechanics is governed by the Schrödinger equation:

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H}\psi, \tag{2}$$

where $\hat{H}$ is the Hamiltonian. For time-independent $\hat{H}$, equation (2) integrates to

$$\psi(t) = e^{-i\hat{H}t/\hbar}\,\psi(0). \tag{3}$$

2.1.1 Stationary states A *stationary state* is an eigenstate of $\hat{H}$; its time dependence is a pure phase, so every expectation value is constant in time. Run-in headings like this one mark the lowest structural level and are kept out of the table of contents.

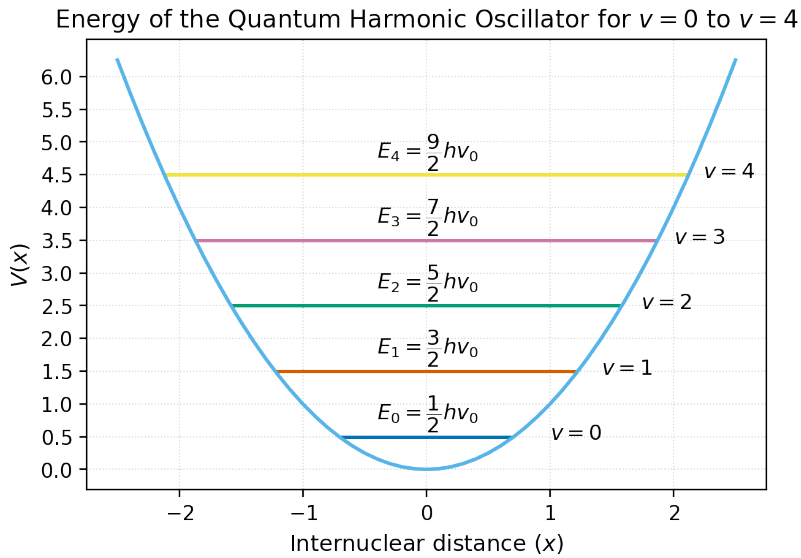

Figure 1. Schematic energy level diagram.

3 Examples

Figure 1 shows a typical structure that recurs throughout the rest of these notes. A few fundamental constants are collected in Table 1 for reference.

Table 1. Fundamental physical constants.

Quantity	Symbol	Value
Speed of light	c	2.998×10^{8} m/s
Planck's constant	$\hbar$	1.055×10^{-34} Js
Elementary charge	e	1.602×10^{-19} C

4 Numerical Methods

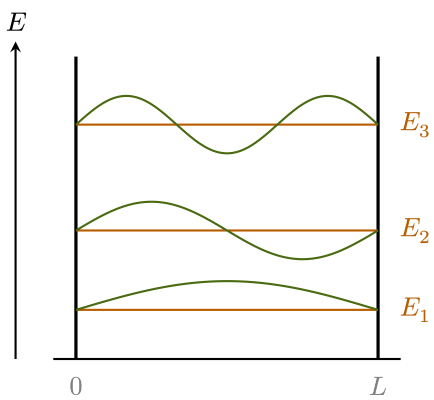

Figure 2. Infinite square well: energy levels $E_n \propto n^2$ (orange) and corresponding eigenfunctions ψ_n (olive).

The `python` environment typesets code with Gruvbox Light syntax highlighting directly in the notes — well suited for short scripts that accompany an analytical derivation. The running example is the infinite square well, shown in Figure 2 with its first three energy levels and eigenfunctions.

The script below evaluates the probability density $|\psi_n(x)|^2$ numerically, then integrates $\langle x\rangle = \int_0^L x\,|\psi_n|^2\,\mathrm{d}x$ for the first three levels and confirms the result $\langle x\rangle = L/2$.

```python
import numpy as np

def psi(n, x, L=1.0):
    """Normalized wavefunction for the infinite square well."""
    return np.sqrt(2.0 / L) * np.sin(n * np.pi * x / L)

def energy(n, L=1.0, hbar=1.0, m=1.0):
    """Energy eigenvalue E_n in natural units."""
    return (n * np.pi * hbar)**2 / (2 * m * L**2)

x = np.linspace(0, 1, 1000)
for n in range(1, 4):
    prob = psi(n, x)**2          # probability density |psi_n|^2
    mean_x = np.trapz(x * prob, x)
    print(f"n={n}  E_n={energy(n):.4f}  <x>={mean_x:.4f}")
```

Running this prints $\langle x\rangle \approx 0.5000$ for all n, consistent with the symmetry of each eigenstate about the midpoint of the well.

5 Exercises

Notes can carry problems alongside the exposition. The `exercise` environment is numbered per section and set at full width, separated from the surrounding text by space alone — less before than after, tying the exercise to its lead-in. `\separator` can additionally mark a change of topic, as below. Subproblems use the `subproblems` environment, which labels parts a), b), c) without affecting ordinary `enumerate` lists.

Exercise 5.1 - Eigenvalues and orthogonality

Show that the eigenvalues of a self-adjoint operator $\hat{A}$ are real, and that eigenvectors belonging to distinct eigenvalues are orthogonal.

Exercise 5.2 - Infinite square well

Consider a particle in a one-dimensional infinite square well of width L.

a) Write down the normalized energy eigenfunctions.

b) Compute the energy levels E_n.

The remaining parts build on the spectrum found above; use the result from part (b) when sketching the densities.

c) Sketch the probability density for $n = 1, 2, 3$.

d) Compute the expectation value $\langle x \rangle$ in the ground state.

6 Closing

These notes are a template; the content above exists only to demonstrate typography. Replace it with your own material as the notes develop.

References

[1] Firstname Lastname. “Example Article Title”. In: *Journal Name* 12.3 (2025), pp. 45–67.

[2] Firstname Lastname. *Example Book Title*. City: Publisher Name, 2025.